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AN ESTIMATE OF THE DYNAMIC EFFECTS OF ECONOMIC INTEGRATION

Josef C. Brada and José A. Méndez*

Abstract—This paper examines two dynamic effects of regional economic integration, higher levels of investment and faster factor productivity growth, in six integration schemes, the EEC, EFTA, LAFTA, CACM, EACM and CMEA. Faster productivity effects are found only in LAFTA and CMEA while all schemes except the CMEA increased members' investment levels. Nevertheless, the cumulative impact of these dynamic effects over nearly twenty years is no more than 1% of members' GNP. Consequently, the dynamic effects of integration cannot explain rapid rates of growth experienced by members of some integration schemes nor can they serve as a rationale for integration.

I. Introduction

The rapid rates of growth achieved by the member countries of the EEC and EFTA in the 1960s created a belief that economic integration has an important effect on the level and growth of economic activity. At first these benefits of integration were attributed to the so-called static effects brought about by the elimination of trade barriers among member countries (Viner (1950), Lipsey (1960), Meade (1968)). Unfortunately, attempts to quantify the static gains from integration revealed these to be disappointingly small.¹ Thus, to explain the evident success of the EEC and EFTA and to serve as the basis for the promotion of economic integration among developing countries, proponents turned to the so-called dynamic effects of integration.² These are the

effects on economic growth due to higher levels of investment and more rapid technological progress.

In this study we estimate the effects of integration on capital formation and on technological progress in six regional integration schemes. Among these are three schemes made up of developed countries, the European Free Trade Area (EFTA), the European Economic Community (EEC), and the Council for Mutual Economic Assistance (CMEA). There are also three developing country schemes, the Central American Common Market (CACM), the Latin American Free Trade Area (LAFTA) and the East African Common Market (EACM). We find that for all integration schemes except the CMEA the level of investment did increase as the result of integration. Increases in technological progress were only evident in LAFTA and CMEA. More important, even though some dynamic effects of integration were evident for each integration scheme, their cumulative impact on the GNPs of the member countries was very small. The largest gain was achieved by the member countries of LAFTA for whom these dynamic effects, cumulated over the period 1960–77, resulted in 1977 GNPs 1.09% higher than they would have been without integration. For the other integration schemes, the comparable gains in members' GNPs were less than 0.5%. Thus, we conclude that while dynamic effects of integration do exist, they have a relatively insignificant impact on the growth rates of member countries' output.

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¹Lipsey (1960) and Johnson (1958) provide the highest estimate of up to 1.0% of GNP in the case of the United Kingdom's membership in the Common Market. More pessimistic estimates are those of Scitovsky (1958), Janssen (1961), and Wemelsfelder (1960).

²The view that the gains from integration were largely dynamic was first put forward by Scitovsky (1958). Dell (1959), Balassa (1961), Thorbecke (1963), and Kreinin (1964) present detailed discussions of these dynamic effects. More recently, Easton and Grubel (1982) argue that the dynamic benefits from tariff reductions are likely to be very large relative to the static effects, while Krueger (1980, 1985) argues that the growth-enhancing effects of more liberal trade regimes are significant for developing countries.

II. The Dynamic Effects of Integration

Unlike the static gains from integration, whose theoretical treatment is well established, the dynamic effects of integration are generally presented as a series of separate and often unrelated phenomena not easily captured by a single model. In this section we attempt to group the dynamic effects of integration into two broad categories. First, integration increases the rate of growth of factor inputs, leading to more rapid growth of output. Second, the rate of technological progress within the economic union increases so that, even if input growth remains unchanged, output nevertheless will grow more rapidly.

Integration is assumed to increase investment by reducing the risk and raising the return to capital. The creation of a large multinational market reduces the risk or raises the expected return to an investment project by enabling firms to lower costs through economies of

scale, by providing a larger pool of customers with heterogeneous tastes, and by enabling the firm to serve national markets with asynchronous business cycles and seasonal buying patterns. Risks of changes in trade partners' commercial policies are reduced by the formation of the integration scheme and investors may experience reduced volatility in share prices due to the integration of national capital markets. Firms will also benefit from factor mobility and from the ability to establish operations in other member countries.

While these effects should ultimately result in a higher level of investment, there may be short-run transitory effects that reduce the volume of investment because some firms will prove unable to compete and begin to disinvest. Others, able to compete within the entire region, may have excess capacity and thus serve the new demand first by operating existing plants more intensively and only later by increasing their investment outlays. Consequently, we model the effects of integration on member-country investment as a long-term effect that shifts the investment function upward, and a transitory effect that decays away over time.

Even if integration did not lead to higher rates of growth of inputs, the growth rate of output could increase because integration promotes a higher level of "disembodied" technological progress among member countries. In part, faster technological progress is brought about by increased competition between firms in member countries, as national monopolies and oligopolies face foreign competitors. This increased competition stimulates research and development, the adoption of new technology and better management practice (Corden (1970) and Martin (1978)). Firm size will increase, permitting greater specialization of production, greater research outlays and economies of scale. Finally, because there is a wider scope for entrepreneurial activity, resources move from stagnant to progressive sectors more rapidly.

This second set of effects suggests that within a production function framework the growth of total factor productivity should increase as a result of integration. Consequently, the disparate sources of dynamic gains from integration can be quantified as an upward shift in the investment function and a higher rate of growth of total factor productivity.

III. Estimation

Because there are two separate effects of integration to be measured, we specify an investment function able to capture the effects of integration on investment behavior and a production function that permits integration to influence the rate of technological progress.

In both developed and developing countries, investment was modelled by means of a flexible accelerator model in which the desired stock of capital is assumed

to be proportional to the expected or planned level of future output.³ Thus, the equation estimated for developing countries is

$$I/Y_{i,t} = a_0 + a_1 \Delta Y/Y_{i,t} + a_2 F/Y_{i,t} + a_3 CU_{i,t} + a_4 CU/t_{i,t} + \sum_i a_{4+i} COUN_i + e_{i,t} \quad (1a)$$

and for developed countries, the equation is

$$I/Y_{i,t} = a_0 + a_1 \Delta Y/Y_{i,t} + a_3 CU_{i,t} + a_4 CU/t_{i,t} + \sum_i a_{4+i} COUN_i + e_{i,t} \quad (1b)$$

where

$I/Y_{i,t}$ = real gross domestic capital formation (I) divided by real gross domestic product (Y) in country i in year t ,

$\Delta Y/Y_{i,t}$ = growth of real gross domestic product in country i in year t ,

$F/Y_{i,t}$ = real foreign capital inflow divided by real gross domestic product in country i in year t ,

$CU_{i,t}$ = 0 if country i was not a member of the integration scheme in year t ,
= 1 otherwise,

$CU/t_{i,t} = (CU_{i,t}/(t - 1950))$,

$COUN_i$ = 1 whenever its observation pertains to country i ,
= 0 otherwise,

$e_{i,t}$ = disturbance term.

The equations have been divided by Y to adjust for heteroscedasticity, and separate country dummies, $COUN_i$, have been introduced to capture any differences that are attributable to individual countries. In addition, in the equation for developing countries, real foreign capital inflows have been added because they have been shown to be an important determinant of investment flows.

The dummy variable $CU_{i,t}$ measures the permanent or long term effect of integration on capital formation. A priori we expect a_3 to be positive. However, as mentioned above, integration may have some transitory effects on the volume of investment, either depressing it below its long-term level at the onset of integration or temporarily raising it above its long-term level. This transitory effect is measured by $CU/t_{i,t}$, a variable that decreases over time. A significant value for a_3 indicates that integration has a long-term impact on the level of

³ Recent studies have shown that this specification performs well in both developed and developing countries; for example, see Clark (1979) for a discussion of investment models applied to the United States, and Blejer and Khan (1984), and the references cited therein, for a treatment of investment in developing countries. The major alternative specification, based on the cost of capital approach of Hall and Jorgenson (1971), would have been very costly to implement for a cross-section sample including developing countries.

capital formation in the integrating countries, while a significant value for a_4 indicates that the short-term impact of integration on capital formation is different from the long-term effect.

Output growth in both developed and developing countries was modelled as depending on the growth of labor and capital inputs, on total factor productivity growth, and on the effect of integration on productivity growth. Thus,

$$\Delta Y/Y_{i,t} = b_0 + b_1 \Delta K/K_{i,t} + b_2 \Delta L/L_{i,t} + b_3 CU_{i,t} + \sum_i b_{4+i} COUN_i + u_{i,t} \quad (2)$$

where

$\Delta K/K_{i,t}$ = rate of growth of capital stock in country i in year t ,

$\Delta L/L_{i,t}$ = rate of growth of population in country i in year t ,⁴

$u_{i,t}$ = disturbance term.

If b_3 is positive, integration increases the rate of technological progress among member countries.

IV. Empirical Results

Data for the period 1951–77 were collected for the members of three developing country schemes, the Central American Common Market (CACM), the Latin American Free Trade Area (LAFTA), and the East African Common Market (EACM) as well as for three developed country schemes, the European Free Trade Area (EFTA), the European Economic Community (EEC) and the Council for Mutual Economic Assistance (CMEA).⁵ Equations (1a), (1b), and (2) were estimated using ordinary least squares (OLS) for each integration scheme by pooling observations across member countries and over years.

OLS parameter estimates for equations (1a) and (1b) are reported in table 1.⁶ The investment equations fit the data well and the model and its estimates generally conform to a priori expectations. In all cases the R^2 s are high, particularly for pooled cross-section data, and the F -values are significant at the 0.0001% level. When significant, estimates of a_1 are, as expected, positive, indicating that capital formation is related to the growth

of output. Also as expected, estimates of a_2 are positive and significant for all developing country integration schemes, indicating that capital formation is related to inflows of foreign capital. All estimates of a_3 are significant at the 5% level and, with the exception of the CMEA, are also positive. Thus integration raised the proportion of output devoted to capital formation both in the case of developing and developed country schemes. In addition, the estimates of a_3 , excluding that of the CMEA, are rather tightly grouped, ranging from 0.020 to 0.049. Thus, among market economies integration has a positive long-term impact on the rate of growth of capital in integrating countries.⁷

The transitory effect of integration on investment, captured by a_4 , is significant at the 5% level only in the case of the CACM and EEC. In the former it is negative and in the latter it is positive. Thus for the CACM, the level of investment in the years immediately following integration actually declined and it was not until the fourth year of integration that the investment effects of integration became positive. In addition, a_4 is negative at lower levels of statistical significance for all other developing country integration schemes. Thus, the negative short-term effects of integration on investment appear to be common in developing country integration schemes. In the case of the EEC, integration led to an initial rise in transitory investment that then tapered off to the new long-term level. It is of some interest to note that, in contrast to the developing country results, the coefficients for the other developed country schemes also indicate that integration initially increased investment above its long-term level. Differences between the experience of developed and developing countries in this regard may very well reflect differences in the effectiveness of integration, in its reciprocity among member countries, and in the ability of member countries' economies to generate the type of phenomena associated with the dynamic effects of integration.

The effects of increased levels of capital formation on output as well as the effects of integration on technological progress are determined by means of equation (2). The OLS parameter estimates are reported in table 2.⁸ The equations' F -values and R^2 s are comparable to those found in Sommers and Suits (1971) and Chenery and Syrquin (1975). Estimates of b_1 , the capital stock

⁴ It is traditional in the empirical growth literature to employ population as a proxy for labor due in part to the unavailability of employment data and because it may be a more meaningful representation of the available labor force.

⁵ A data appendix is available from the authors upon request.

⁶ To conserve space, the estimates for the country dummies are omitted from the table. The null hypothesis that

$$b_4 = \dots = b_{4+i} = \dots = b_{4+k} = 0$$

where k = the number of countries in a particular integration scheme was rejected for the investment equations.

⁷ The negative a_3 in the case of CMEA may be explained in two ways. First it may reflect the discontinuation of the Stalinist investment policies of the early 1950s. Secondly, it should be recognized that one of the goals of CMEA integration was to reduce the investment burden on the member countries by rationalizing investment decisions. Thus we should not expect to see in CMEA the same increase in investment that is evident in integration among market economies.

⁸ The country dummies were dropped from equation (2) for each integration scheme because the (joint) null hypothesis that they do not contribute to an explanation of capital formation could not be rejected.

TABLE 1.—OLS REGRESSION ESTIMATES FOR EQUATION (1)
DEPENDENT VARIABLE: I/Y

Integration Scheme	a_0	$\Delta Y/Y$ a_1	F/Y a_2	CU a_3	CU/t a_4	R^2	Standard Error	F-ratio (Prob > F)	Number of Observations
CACM	0.118 (24.39)	0.187 (3.43)	0.690 (7.60)	0.032 (3.31) ^a	-0.417 (-2.61) ^a	.630	.0246	25.71 (.0001)	130
LAFTA	0.148 (44.01)	-0.020 (-1.22)	0.458 (8.47)	0.024 (3.54) ^a	-0.168 (-1.51)	.611	.0283	29.42 (.0001)	227
EACM	0.143 (26.70)	-0.025 (-0.62)	0.319 (3.20)	0.049 (2.48) ^a	-0.466 (-1.11)	.521	.0312	9.62 (.0001)	60
EEC	0.203 (52.47)	-0.055 (-1.26)	—	0.020 (3.24) ^a	0.208 (2.33) ^a	.550	.0197	22.43 (.0001)	156
EFTA	0.194 (56.10)	-0.057 (-1.11)	—	0.035 (5.96) ^a	0.094 (1.04)	.837	.0199	97.77 (.0001)	182
CMEA	0.448 (24.92)	0.186 (0.88)	—	-0.239 (-5.12) ^a	0.517 (0.56)	.813	.1326	81.79 (.0001)	179

Notes: All equations estimated with constraint: $\sum_i a_{4+i} COUN_i = 0$. t -ratios are in parentheses.^aCoefficient estimate of CU or CU/t significant at the 5% level.

coefficient, are significant for all integration schemes, whereas b_2 , the coefficient for the rate of growth of the population, is not significant except for the EFTA. For developing countries where labor is in surplus, such a conclusion is not surprising; nor is the negative b_2 for the EACM where workers may have been redundant. The coefficient for the integration dummy, b_3 , is positive and significant for LAFTA and the CMEA at the 10% level of significance, indicating that the rate of growth of total factor productivity increased following integration. However, b_3 is negative and significant for the EFTA at the 5% level indicating that integration reduced the rate of productivity growth.

V. Measuring the Gains from Integration: Implications for Integrating Countries

Having thus demonstrated that economic integration has some impact on total factor productivity growth

and on the rate of capital formation in integrating countries, we investigate whether such gains are of sufficient magnitude to make economic integration an important mechanism for promoting growth. To do this we compare the GDP that the member countries could have achieved either at the time the integration scheme was terminated or in 1977, the last year of our sample, with and without the dynamic benefits of integration. Terminal year GDPs without integration were computed by means of a dynamic simulation of equations (1a) or (1b), and (2) from the first year of the sample to the terminal year with $CU_{i,t} = 0$ for all years and all countries.⁹ Terminal year GDPs with dynamic effects of integration were also calculated by means of the same equations, but with $CU_{i,t} = 1$ for those years in which

⁹Equations (1a) and (1b) were post-multiplied by Y_t/K_{t-1} and then substituted for the variable $\Delta K/K$ in equation (2).

TABLE 2.—OLS REGRESSION ESTIMATES FOR EQUATION (2)
DEPENDENT VARIABLE: $\Delta Y/Y$

Integration Scheme	b_0	$\Delta K/K$ b_1	$\Delta L/L$ b_2	CU b_3	R^2	Standard Error	F-ratio (Prob > F)	Number of Observations
CACM	-0.023 (-1.18)	1.290 (4.58)	0.009 (0.02)	0.011 (1.50)	.202	.0393	10.62 (.0001)	130
LAFTA	-0.028 (-1.18)	0.935 (2.65)	0.590 (0.75)	0.023 (1.63) ^b	.054	.1063	5.21 (.0018)	277
EACM	-0.025 (-0.38)	1.952 (2.19)	-1.232 (-0.96)	-0.019 (-0.53)	.095	.1063	1.97 (.0954)	60
EEC	-0.012 (-0.69)	1.073 (3.03)	0.979 (1.10)	-0.007 (-0.94)	.096	.0378	5.39 (.0001)	156
EFTA	-0.005 (-0.51)	1.089 (5.17)	0.427 (4.74)	-0.011 (-2.20) ^a	.133	.0287	9.13 (.0001)	182
CMEA	-0.009 (-0.53)	0.576 (2.21) ^a	0.122 (0.24)	0.019 (1.84) ^b	.031	.0494	1.86 (.1370)	179

Note: t -ratios are in parentheses.^aCoefficient estimate of CU significant at the 5% level.^bCoefficient estimate of CU significant at the 10% level.

TABLE 3.—INCREASE IN TERMINAL-YEAR GROSS DOMESTIC PRODUCT DUE TO DYNAMIC EFFECTS OF INTEGRATION (AS PERCENTAGES OF TERMINAL YEAR GDP WITHOUT INTEGRATION)

Integration Scheme	Increase in GDP Due to Higher Investment Level	Increase in GDP Due to Higher Rate of Growth in Factor Productivity	Total = 1 + 2
CACM 1961–77 ^a	0.35	NS	0.35
LAFTA 1960–77 ^b	0.18	0.91	1.09
EACM 1967–72	0.18	NS	0.18
EEC 1959–77 ^c	0.48	NS	0.48
EFTA 1960–77 ^d	0.32	–0.35	–0.03
CMEA 1964–77	–0.87	0.76	–0.11

Note: NS = Technological progress coefficient not significant at 10% level.

^aExcept Honduras 1961–69; Nicaragua, 1962–; Costa Rica, 1963–.

^bExcept Colombia and Ecuador, 1961–; Venezuela, 1965–; Bolivia 1967–.

^cExcludes Denmark and United Kingdom, 1973–.

^dIncludes Denmark and United Kingdom through 1977.

each individual country belonged to the integration scheme. Only those values of a_3 , a_4 , and b_3 that were significantly different from zero at the 10% significance level were employed in these calculations.

The results are reported in table 3. Although terminal-year GDP was calculated on a country-by-country basis, for brevity we report the comparisons only by integration schemes. LAFTA, by virtue of significant effects on both capital formation and productivity growth achieved the largest gain, with terminal year GDPs 1.09% higher than they would have been without integration. For the other integration schemes gains did not exceed a 0.5% gain in terminal year GDPs over their no-integration levels. In view of the fact that these gains reflect the cumulative effects of nearly two decades of integration, such modest gains indicate that the dynamic effects of integration are likely to be of minor consequence to countries contemplating participation in such schemes.

The effects of integration on productivity growth, where they are significant, are considerably larger in absolute value than the investment effects. Thus LAFTA's superior results are largely the result of a large and significant increase in factor productivity. The productivity effect for CMEA is also larger than the investment effect for any of the other integration schemes. Unfortunately, gains in productivity growth are evident in only two of the integration schemes, and thus countries entering into integration schemes cannot be certain that they will be able to reap this, the quantitatively more important, dynamic effect. On the other hand, integrating countries can expect with some certainty a long-term increase in their level of investment. This increase, although statistically significant, makes only a minor contribution to economic growth. For developing countries it is also possible that there will be transitory effects of integration that will temporarily reduce the volume of investment.

Based on our examination of six integration schemes, we thus conclude that, while the existence of dynamic effects of integration can be demonstrated, the impact of these effects on economic growth is quite modest. Dynamic effects of integration can neither explain the rapid growth of West European economies in the 1960s nor serve as a strong justification for encouraging non-member countries to join existing schemes or to organize new ones.

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CREATIVE DESTRUCTION AND THE BEHAVIOR OF PRODUCTIVITY OVER THE BUSINESS CYCLE

Edward Montgomery and William Wascher*

Abstract—We develop and test a model of short-run variations in productivity which explicitly incorporates the Schumpeterian forces of "creative destruction." Using data on business failures and formations, we find that increases in competitive pressures reduce the extent of labor hoarding and raise the level of productivity in the economy.

I. Introduction

Schumpeter's (1934, 1939) theory of innovations has long been used to support the notion of long waves of economic growth or stagnation. The purpose of this paper is to apply a part of Schumpeter's theory to an analysis of movements in productivity over the business cycle. In particular, we develop and test a model of short-run variations in productivity that explicitly incorporates the forces underlying "creative destruction": the birth and death of firms due to the diffusion of new technology into the production process.

Previous research on the connection between the theory of innovation and productivity has concentrated on its implications for long-run aggregate productivity growth.¹ Empirically, the trend rate of growth of productivity has been found to change over time as increases in competitive pressure raise the rate of technological progress in the economy. The notion that changes

in competitive pressures may influence productivity also may be relevant in a cyclical model of productivity. If the extent of competitive pressure differs across business cycles or at various stages over the business cycle, productivity patterns may vary. In particular, when competitive pressures are greater, firms may be induced to adjust their stocks of labor inputs more quickly in response to changes in production. At an aggregate level, the result would be a more rapid cyclical response of total hours to movements in aggregate demand and, thus, smaller fluctuations in measured average labor productivity over the business cycle.

In addition, changes in the mix of efficient and inefficient firms associated with the exit and entry of firms into the marketplace may influence patterns of productivity growth. If existing firms are less efficient and entering firms are more efficient, then productivity should rise when either business failures or new entrepreneurial activity are at high levels.

In this paper, we incorporate the effects associated with creative destruction into a stock adjustment model of productivity using data on business failures and formations. On the whole, the results statistically support the relevance of the Schumpeterian model to cyclical productivity behavior. Changes in the rate of new business formations and business failures are found to affect productivity by altering both the mix of firms and the extent of labor hoarding.

II. A Cyclical Model of Productivity

Following Fair (1985), Clark (1984), Gordon (1979), and others, we model short-run fluctuations in productivity as resulting from labor hoarding behavior on the part of firms. That is, because of hiring and training costs, labor is a quasi-fixed factor of production whose

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¹ Weisskopf, Bowles, and Gordon (1983) and Gay and Roach (1986) are recent examples.